

Problem Solving 1

* Common elements

$$\checkmark A = \{1, 2, 2, 3, 4\}$$

$$Ans = \underline{\underline{2, 2, 3}}$$

$$B = \{2, 2, 2, 3, 3\}$$

→

HM

$$\left\{ \begin{array}{l} 1 \rightarrow 1 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \\ 4 \rightarrow 1 \end{array} \right\}$$

* Count subarr with zero sum

$$or(10) = \begin{array}{|c|c|c|c|c|c|c|c|c|c|c|} \hline 2 & 2 & 1 & -3 & 4 & 3 & 1 & -2 & -3 & 2 & -2 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ \hline \end{array}$$

$$Pf(10) = \begin{array}{|c|c|c|c|c|c|c|c|c|c|c|} \hline 2 & 4 & 5 & 2 & 6 & 9 & 10 & 8 & 5 & 7 & 5 & 0 \\ \hline 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ \hline \end{array}$$



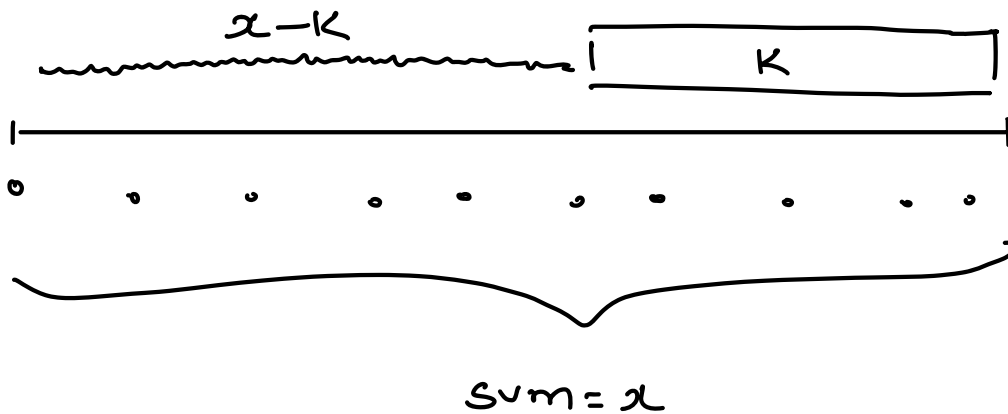
sum	HM	count
0	→	1
2	→	+2
4	→	1
5	→	$\times 2$
6	→	1
9	→	1
10	→	1
8	→	1
7	→	1

0

$$ans = 0 + 1 + 1 + \underline{\underline{2}}$$

↓
hm.get(s)

* subarray sum equals K



subarr sum = 1

$$arr[10] =$$

2	2	1	-3	4	3	1	-2	-3	2
0	1	2	3	4	5	6	7	8	9

$A[i]$

$k = 0$
0 - _

$Pf[10]$

2	4	5	2	6	9	10	8	5	7
0	1	2	3	4	5	6	7	8	9

$k = 7$

10 | 1

HM = {

Sum	freq
0	→ 1
2	→ 2
4	→ 1
5	→ 1
6	→ 1

}

Subarr sum

Subarr sum

Sum - k

$$2 - 7 = -5$$

$$4 - 7 = -3$$

$$5 - 7 = -2$$

$$2 - 7 = -5$$

$$6 - 7 = -1$$

$$9 - 7 = 2$$

* Pair diff

$$x - y = k$$

$A[i]$

$$y = \underline{\underline{x - k}}$$

$$y = x + k$$

$$\text{diff} = 5$$

5

10

15